EAT: How Far Do Raw pLM Embeddings Carry Annotations?

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Most proteins will never be measured

150,000,000

protein sequences in UniProtKB [1]

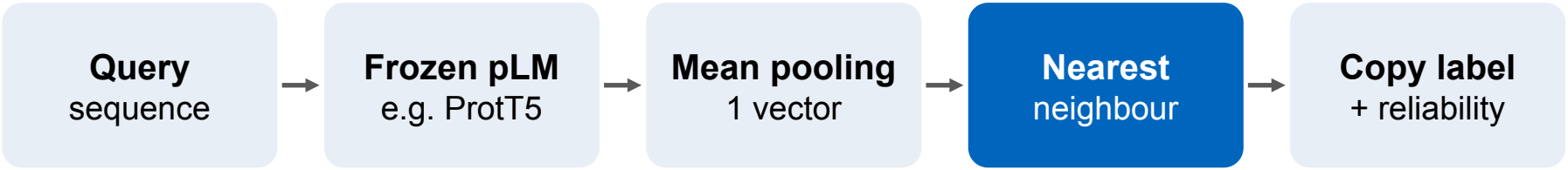
48,401

with experimentally measured enzyme activity [1]

→ **Everything else is annotated by transfer from a similar protein.**
Sequence similarity (BLAST, MMseqs2) fails below ~30 % identity.

[1] UniProt Consortium. UniProt: the Universal Protein Knowledgebase. Nucleic Acids Res (2025); counts: UniProtKB 2026_03, queried 28.10.2026.

EAT: a query inherits its nearest neighbour's label



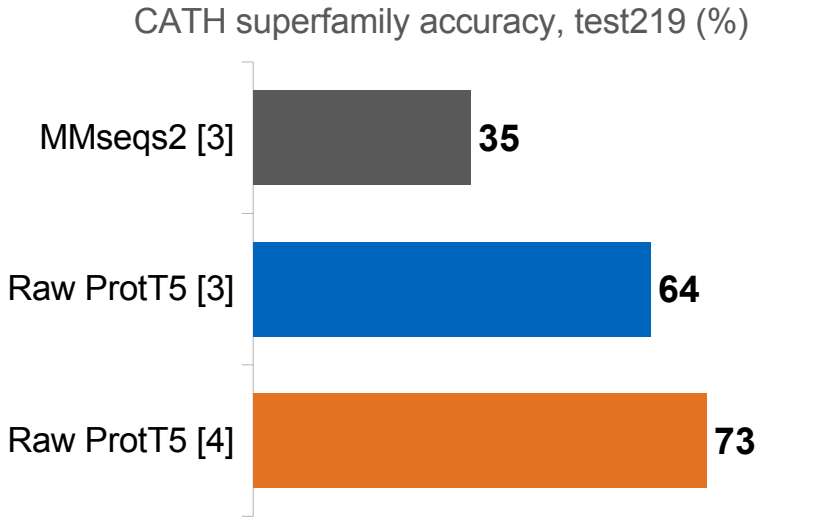
Reliability index (goPredSim): **RI = 0.5 / (0.5 + d)**

Training-free, yet beats BLAST for GO terms [2]

CATH superfamily, remote homologues: raw ProtT5 64 % vs MMseqs2 35 % [3]

[2] M. Littmann … B. Rost. Embeddings transfer GO annotations beyond homology. Sci Rep (2021).
[3] M. Heinzinger … B. Rost. Contrastive learning on protein embeddings enlightens midnight zone. NAR Genom Bioinform (2022).

Raw EAT: a common baseline, set up differently each time



Same pLM, same test + lookup sets: 9 points apart. No CIs reported.

- Every paper picks pLM, pooling, distance, k and lookup set differently
- The pLM matters: ProtT5 → ProstT5 raises CATH-H from 69.6 to 78.5 % [5]
- RI cutoffs were tuned for SeqVec (2021); distance scales differ across pLMs
- **No "HFSP for embeddings" [6]:** no calibrated safe-transfer threshold

[3] Heinzinger … Rost (2022). [4] A. Elnaggar … B. Rost. Ankh: Optimized protein language model. arXiv (2023).
[5] D. Miller … C. Orengo. ContrasTED. bioRxiv (2026). [6] Y. Mahlich … Y. Bromberg. HFSP. Bioinformatics (2018).

Two research questions

RQ1 Which pLM and distance?
Which pLM and distance (Euclidean vs cosine) transfer structure (CATH) and function (EC) best – and where does EAT beat sequence similarity?

RQ2 When is a transfer safe?
Up to which distance is transfer safe (e.g. ≥ 95 % precision)? One reliability score that means the same for every pLM (0.9 = 90 % correct).

What “raw” means

- Frozen pLM · mean of last-layer residue embeddings · label of nearest neighbour(s)
- Nothing trained on top: no contrastive learning, probes or fine-tuning
- Labels only on a calibration set – never on the test set

11 pLMs × 2 distances × 2 tasks, each vs sequence search

Family	Models	Dim
ProtT5	ProtT5-XL-U50 · ProstT5	1024
ESM-2	8M · 35M · 150M · 650M · 3B	320 → 2560
ESM C	300M · 600M	960 · 1152
Ankh	base · large · Ankh3-large	768 · 1536

Distance
Euclidean · cosine

Tasks
CATH C/A/T/H · EC 1–4

Baseline
best MMseqs2 hit

Accuracy per level with confidence intervals, binned by sequence identity to the lookup set

Embedding sizes: Hugging Face model configs; ESM C from the EvolutionaryScale ESM repository.

Data: CATH is ready-made, the EC set we must build

CATH (structure)

- ProtTucker benchmark: test219 vs lookup69k [3]
- Then non-redundant CATH v4.4 sets
- Pitfall: CATH-S100 lookup contains test219

EC (function)

- No ready-made split
- UniProt: 48,401 entries with experimental activity
- HFSP set: 7,022 enzymes [6]

Embeddings

- ProtSpace notebook (all but ESM-2 3B)
- ESM-2 3B via HF transformers
- EC lookup = compute bottleneck

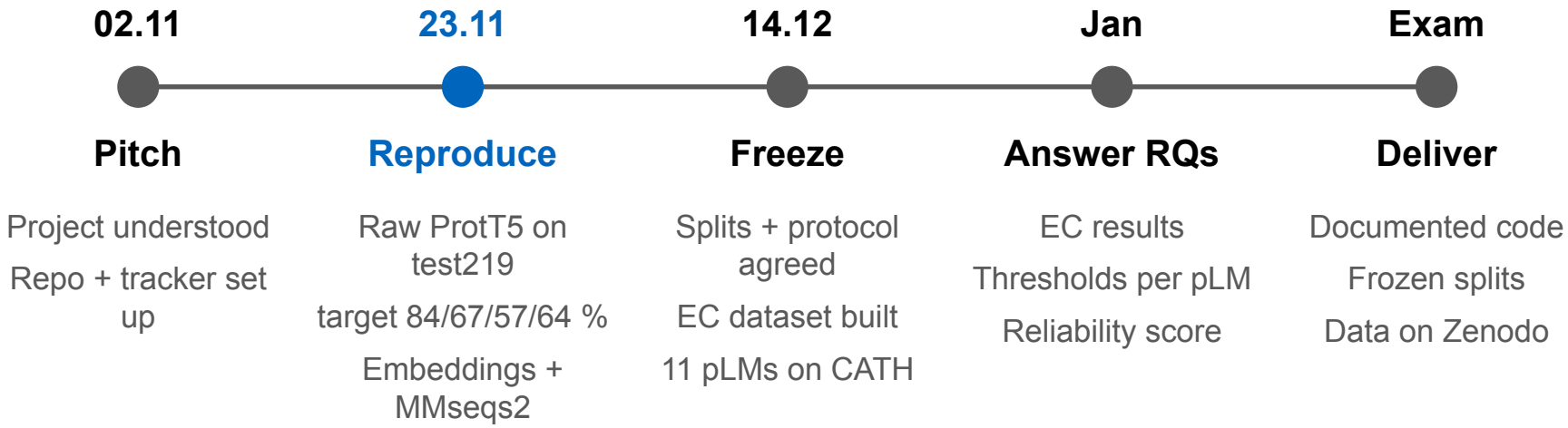
[3] Heinzinger … Rost (2022). [6] Mahlich … Bromberg (2018). CATH v4.4: download.cathdb.info. EAT code: github.com/Rostlab/EAT

Protocol: tune on calibration data, report on test data

Lookup	**Calibration**	**Test**
labelled proteins to transfer from	choose settings, fit thresholds + score	touched once: final numbers

1 Accuracy	per level, bootstrap 95 % CI, binned by sequence identity, vs MMseqs2
2 Safe distance	largest d with precision ≥ 95 % on calibration → checked on test
3 Reliability	distance → calibrated probability (e.g. isotonic), same meaning for every pLM

Plan: reproduce first, then freeze the protocol, then scale



Design questions we want to settle with you

Splits	How similar may queries be to the lookup set?	idea: *cap identity, report per identity bin*
EC labels	Which EC numbers can we trust as test labels?	idea: *only reactions with experimental evidence*
Model size	Bigger ESM-2: more knowledge or more dimensions?	idea: *compare at matched dimension (PCA)*
Distances	When do Euclidean and cosine disagree?	idea: *only via vector norms; compare calibrated precision*
Edge cases	Multi-domain proteins, several / incomplete ECs	idea: *define per-level correctness up front*

Status and what we need

Done

- Project + key papers read
- Repo, issue tracker, protocols
- Plan + protocol draft

Expected roadblock

- Compute: EC lookup set
- ESM-2 3B embeddings
- k-NN search at scale

Asks for you

- GPU / cluster access?
- EC source: UniProt or HFSP?
- When to freeze splits?

References

[1] UniProt Consortium. UniProt: the Universal Protein Knowledgebase. Nucleic Acids Res (2025).

[2] M. Littmann … B. Rost. Embeddings from deep learning transfer GO annotations beyond homology. Sci Rep (2021).

[3] M. Heinzinger … B. Rost. Contrastive learning on protein embeddings enlightens midnight zone. NAR Genom Bioinform (2022).

[4] A. Elnaggar … B. Rost. Ankh: Optimized protein language model unlocks general-purpose modelling. arXiv (2023).

[5] D. Miller … C. Orengo. ContrasTED: Contrastive domain embeddings for remote homology classification. bioRxiv (2026).

[6] Y. Mahlich … Y. Bromberg. HFSP: High speed homology-driven function annotation of proteins. Bioinformatics (2018).

Project description: Chair of Bioinformatics (I12), TUM. EAT: A benchmark for annotation transfer with raw pLM embeddings (2026).